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Growth properties of Fourier transforms via moduli of continuity

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To Paul Malliavin, friend and teacher

Abstract

We obtain new inequalities for the Fourier transform, both on Euclidean space, and on non-compact, rank one symmetric spaces. In both cases these are expressed as a gauge on the size of the transform in terms of a suitable integral modulus of continuity of the function. In all settings, the results present a natural corollary: a quantitative form of the Riemann–Lebesgue lemma. A prototype is given in one-dimensional Fourier analysis.
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1. Introduction and motivation

A classical theme in Fourier analysis is embodied in the statement: behavior of the modulus of continuity of a function for small parameter reflects in the behavior of the Fourier transform for large parameter. Perhaps the simplest example of this principle is the following estimate, due to Lebesgue. Let $\omega_1[f](r) = \sup_{|u|\leqslant r} \frac{1}{4\pi} \int_{-\pi}^{\pi} |f(t+u) - f(t)|\, dt$ be the L^1-modulus of continuity

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on the circle group $(-\pi, \pi]$, then the Fourier coefficients of an L^1 function f satisfy

$$|\widehat{f}(n)| \leqslant \frac{1}{4\pi} \int_{-\pi}^{\pi} \left| f\left(t + \frac{\pi}{n}\right) - f(t) \right| dt \leqslant \omega_1[f]\left(\frac{\pi}{|n|}\right). \tag{1.1}$$

This simple inequality immediately implies the Riemann–Lebesgue lemma ($\widehat{f}(n) = o(1)$, $|n| \to \infty$), and in a sense is a better result, providing a quantitative gauge on the size of the Fourier coefficients in terms of the L^1-modulus of continuity. Numerous other examples of this principle are available, e.g. Zygmund [23], Titchmarsh [21], and Stein [19].

The focus of this paper is to obtain weighted L^p-estimates of the Fourier transform of a function in terms of an integral modulus of continuity defined via spherical means. The spherical mean operator, being a function of the Laplacian, will be seen as a natural choice to use to gauge the size of Fourier transforms. Our viewpoint is carried out in Euclidean space and on rank one symmetric spaces of non-compact type.

In part, our current interest in this theme stems from a result of Gioev [9] which is based on the work of Cline [4]. In order to state this result, a few notations are in order. Let $f : \mathbb{R}^n \to \mathbb{R}$ with mth order differences defined by

$$\Delta_y^m f(x) = \sum_{k=0}^{m} (-1)^{m-k} \binom{m}{k} f(x + ky), \quad m \geqslant 1. \tag{1.2}$$

Let $p, r \geqslant 1$, Cline introduced the following moduli of continuity:

$$\omega_{p,m,r}[f](r) = \left(\int_{S^{n-1}} \left\| \Delta_{r\omega}^m f \right\|_p^r d\omega \right)^{\frac{1}{r}}, \tag{1.3}$$

$$\omega_{p,m,\infty}[f](r) = \sup_{|y| \leqslant r} \left\| \Delta_y^m f \right\|_p. \tag{1.4}$$

Here $d\omega$ is the usual surface measure on the unit sphere S^{n-1} embedded in $\mathbb{R}^n$. Throughout this paper, two conventions are used: (1) the Holder conjugate exponent to p is denoted q, $p^{-1} + q^{-1} = 1$; (2) constants appearing in various estimates are denoted C, c, etc., are not to be interpreted as being the same in every occurrence.

Theorem 1 *(Gioev). Let $n \geqslant 2$ and $1 \leqslant p \leqslant 2$. Then there exists a constant $c = c(p, n, m)$ such that for $f \in L^p(\mathbb{R}^n)$ and $s \geqslant q$,*

$$\left(\int_{\mathbb{R}^n} \min\left\{ 1, \left(\frac{|\xi|}{r} \right)^{mq} \right\} |\widehat{f}(\xi)|^q \, d\xi \right)^{1/q} \leqslant c\omega_{p,m,s}[f]\left(\frac{1}{r}\right)$$

in the case where $p > 1$, and for $p = 1$,

$$\sup_{\xi} \left[\min\left\{ 1, \left(\frac{|\xi|}{r} \right)^{m} \right\} |\widehat{f}(\xi)| \right] \leqslant c\omega_{1,m,\infty}[f]\left(\frac{1}{r}\right).$$

Gioev proved this for the case $s = q$. For $s > q$, the result follows by monotonicity of the modulus of continuity in s. If $s < \infty$, the result is false when $n = 1$ [9]. Intuitively this makes sense: in dimension one, the unit sphere consists of two points and averaging produces no useful cancellation.

In Euclidean space, the Fourier transform of L^p-functions, $1 < p \leqslant 2$, is defined via continuous extension using the Plancherel and Hausdorff–Young theorems. Here our result takes the form of a variation of Gioev's theorem. However, the estimates of the Euclidean spherical function are crucial to obtaining the analogous estimates in the rank one case (see Lemma 9). In the symmetric space setting our results are new and take two forms. One form is based on the fact that the Helgason Fourier transform $\widehat{f}(\lambda, b)$ (see Section 3 for the definition) is defined for L^p-functions, $1 \leqslant p < 2$, as an absolutely convergent integral for λ in a certain strip in the complex plane, a fact observed in [17]. This fact is well-known in the spherical transform case and relates to the classical Kunze–Stein phenomena [8,12,13]. As a corollary, we obtain a quantified form of the Riemann–Lebesgue lemma on rank one symmetric spaces given in [17]. The second form of our results are obtained by an application of the Plancherel theorem and known Hausdorff–Young inequalities for the Helgason Fourier transform. It will also be seen that the ideas presented have natural extension into harmonic analysis associated with the Bessel and Jacobi differential operators.

The following result in one dimension serves as a model case for part of the sequel. We introduce the notation

$$Z_h[f] := f_h(x) + f_{-h}(x) - 2f(x) \tag{1.5}$$

for the symmetric difference, where $f_h(x) := f(x + h)$ denotes the translate of f. The natural L^p-modulus of continuity is

$$\Omega_p[f](t) := \sup_{0 \leqslant h \leqslant t} \left\| Z_h[f] \right\|_p. \tag{1.6}$$

Proposition 2. *Let $1 \leqslant p \leqslant 2$ and $f \in L^p(\mathbb{R})$. Then there exists a constant $c_p > 0$ such that,*

$$\left(\int_{\mathbb{R}} \min\left\{ 1, \left(\frac{\lambda}{r} \right)^{2q} \right\} \left| \widehat{f}(\lambda) \right|^q d\lambda \right)^{1/q} \leqslant c_p \Omega_p[f]\left(\frac{1}{r} \right), \quad 1 < p \leqslant 2,$$

$$\sup_{\lambda} \left[\min\left\{ 1, \left(\frac{\lambda}{r} \right)^2 \right\} \left| \widehat{f}(\lambda) \right| \right] \leqslant c_1 \Omega_1[f]\left(\frac{1}{r} \right), \quad p = 1.$$

Proof. Standard Fourier transform formulas provide the identity

$$\widehat{Z_h[f]}(\lambda) = -2 \sin^2\left(\frac{\lambda h}{2} \right) \widehat{f}(\lambda).$$

Applying the Hausdorff–Young inequality one has

$$\sup_{0\leqslant h\leqslant 1/r}\left\|Z_h[f]\right\|_p^q \geqslant c'_p \sup_{0\leqslant h\leqslant 1/r}\int_{\mathbb{R}} \sin^{2q}\left(\frac{\lambda h}{2}\right)\left|\widehat{f}(\lambda)\right|^q d\lambda$$

$$= c'_p \sup_{0\leqslant z\leqslant 1}\int_{\mathbb{R}} \sin^{2q}\left(\frac{\lambda z}{2r}\right)\left|\widehat{f}(\lambda)\right|^q d\lambda \geqslant c'_p \int_{\mathbb{R}}\left[\int_0^1 \sin^{2q}\left(\frac{\lambda z}{2r}\right)\right]\left|\widehat{f}(\lambda)\right|^q d\lambda.$$

Elementary estimates and computations show that

$$\int_0^1 \sin^{2q}\left(\frac{\lambda z}{2r}\right) dz \geqslant c\min\left(1, \left(\frac{\lambda}{r}\right)^{2q}\right)$$

for a suitable positive constant c. Collecting constants concludes the proof. □

The following corollary is immediate and may be viewed as a quantitative Riemann–Lebesgue lemma.

Corollary 3. *Let* $1 \leqslant p \leqslant 2$ *and* $f \in L^p(\mathbb{R})$. *Then*

$$\left(\int_{|\lambda|\geqslant r}\left|\widehat{f}(\lambda)\right|^q d\lambda\right)^{1/q} \leqslant c_p \Omega_p[f]\left(\frac{1}{r}\right), \quad 1 < p \leqslant 2,$$

$$\sup_{|\lambda|\geqslant r}\left|\widehat{f}(\lambda)\right| \leqslant c_1 \Omega_1[f]\left(\frac{1}{r}\right), \quad p = 1.$$

The Fourier transform on the real line provides the explicit spectral theory of the operator d^2/dx^2. Many authors have extended fundamental ideas of Fourier analysis by recognizing that d^2/dx^2 is a special limiting case of more general differential operators. In particular we focus on the following:

- Let $\alpha > -1$ and consider the differential operator

$$D_\alpha := \frac{d^2}{dx^2} + \frac{2\alpha+1}{x}\frac{d}{dx}. \tag{1.7}$$

The equation $D_\alpha u + \lambda^2 u = 0$ has solutions regular at the origin, given by the spherical Bessel functions

$$j_\alpha(\lambda x) = 2^\alpha \Gamma(\alpha+1)(\lambda x)^{-\alpha} J_\alpha(\lambda x),$$

where J_α is the usual Bessel function of the first kind and we have normalized so that $j_\alpha(0) = 1$. The spectral theory of D_α is provided by the Fourier–Bessel transform

$$\mathcal{F}_\alpha f(\lambda) = \int_0^\infty f(x) j_\alpha(\lambda x) x^{2\alpha+1}\, dx. \tag{1.8}$$

The cosine transform is the special case when $\alpha = -1/2$.

- Let $\alpha > -1/2$, $-1/2 \leqslant \beta \leqslant \alpha$, $\rho = \alpha + \beta + 1$, and consider

$$\Delta_{(\alpha,\beta)}(x) = (2\sinh x)^{2\alpha+1}(2\cosh x)^{2\beta+1},$$

$$D_{(\alpha,\beta)} = \frac{d^2}{dx^2} + \frac{\Delta'_{(\alpha,\beta)}(x)}{\Delta_{(\alpha,\beta)}(x)}\frac{d}{dx}.$$

Smooth solutions of the equation $D_{(\alpha,\beta)}u + (\lambda^2 + \rho^2)u = 0$ which are 1 at $x = 0$ are the Jacobi functions of the first kind,

$$\phi_\lambda^{\alpha,\beta}(x) = F\left(\frac{1}{2}(\rho - i\lambda); \frac{1}{2}(\rho + i\lambda); \alpha + 1; -\sinh^2 x\right).$$

Here, F is the Gauss hypergeometric function. The spectral theory of $D_{(\alpha,\beta)}$ is provided by the Jacobi transform:

$$\mathcal{F}_{(\alpha,\beta)} f(\lambda) = \int_0^\infty f(x)\phi_\lambda^{(\alpha,\beta)}(x)\Delta_{(\alpha,\beta)}(x)\, dx.$$

See Koornwinder [12]. The cosine transformation appears again in the limiting case of $\beta = \alpha = -1/2$.

In order to develop a result akin to Theorem 2 in these settings, a suitable replacement for translation on the real line must be introduced. Such "generalized translation operators" have been introduced and studied, see [22]. In the Jacobi function framework, these were introduced by Flensted-Jensen and Koornwinder [7].

Alternatively, both of the above mentioned operators have direct connection to geometrical settings as follows.

- *Bessel case.* Let $\alpha = (n-2)/2$. Then the operator D_α is the radial part of the Laplacian on $\mathbf{R}^n$ and the associated Fourier–Bessel transform is the Fourier transform on $\mathbf{R}^n$, restricted to radial functions.
- *Jacobi case.* For appropriate choices of parameters (α, β), the operator $D_{\alpha,\beta}$ is the radial part of the Laplacian on a rank one symmetric space of non-compact type. The Jacobi transform is then the Helgason Fourier transform on this symmetric space, restricted to radial functions, also called the *spherical transform*.

As mentioned above, the spherical mean operator is a natural candidate to use in defining an integral modulus of continuity. The spherical mean operator is also known to motivate the definition of the aforementioned generalized translation operators.

In Section 2, our variation of Gioev's result in Euclidean space is developed. The heart of the paper is in Section 3 where the ideas are extended to the rank one symmetric spaces. Section 4 indicates how the ideas can be applied to harmonic analysis of Bessel and Jacobi differential operators. Both the Bessel and Jacobi differential operators belong to a broader class of singular differential operators, e.g., see [22]. It is an interesting problem as to whether our analytical results of Section 4 can be proved in the more general framework.

2. Euclidean space

The Fourier transform on $\mathbf{R}^n$ is defined by

$$\widehat{f}(\xi) := \int_{\mathbf{R}^n} f(x) e^{-ix\cdot\xi}\, dx, \quad f \in L^1\big(\mathbf{R}^n\big). \tag{2.1}$$

Basic references are Stein and Weiss [20] and Pinsky [16]. The Plancherel theorem provides an extension of the Fourier transform to $L^2(\mathbf{R}^n)$. Furthermore, the Hausdorff–Young theorem provides the extension to $L^p(\mathbf{R}^n)$ for $1 < p < 2$, with the bound $\|\widehat{f}\|_q \leqslant c_p \|f\|_p$ where $q = p/(p-1)$ is the conjugate exponent and c_p is a positive constant.

Suppose now that f is a radial function; by abuse of notation, we write $f(x) = f(r)$, $r = |x|$. Then the Fourier transform of f is also radial and we write $\widehat{f}(\xi) = \widehat{f}(\lambda)$. In this case (2.1) can be written

$$\widehat{f}(\lambda) = \omega_{n-1} \int_0^\infty f(r) \left(\frac{1}{\omega_{n-1}} \int_{S^{n-1}} e^{-ir\lambda(\omega\cdot\omega')}\, d\omega' \right) r^{n-1}\, dr. \tag{2.2}$$

Here $\xi = \lambda\omega$, $x = r\omega'$. The function in brackets is precisely the spherical Bessel function $j_{(n-2)/2}$ mentioned in Section 1.

Returning to the case of general functions $f(x)$, the spherical mean operator can be defined by

$$M^t f(x) = \frac{1}{\omega_{n-1}} \int_{S^{n-1}} f(x + t\omega)\, d\omega. \tag{2.3}$$

For $f \in L^p(R^n)$ and $p \geqslant 1$, $M^t f(x)$ is defined a.e. (Lebesgue). Furthermore, $\|M^t f\|_p \leqslant \|f\|_p$ and $\lim_{t\to 0^+} M^t f = f$ a.e. and in L^p-norm. An operational property of the Fourier transform is

$$\widehat{M^t f}(\xi) = \widehat{f}(\xi)\, j_{\frac{n-2}{2}}\big(|\xi|t\big). \tag{2.4}$$

We introduce the spherical modulus of continuity:

$$\Omega_p[f](r) := \sup_{0\leqslant t\leqslant r} \big\| M^t f - f \big\|_p. \tag{2.5}$$

Our variation of Gioev's result is the following:

Proposition 4. *Let $1 \leqslant p \leqslant 2$ and $f \in L^p(R^n)$. Then there exists a constant $C = C(p,n)$ such that*

$$\left(\int_{\mathbf{R}^n} \min\left\{1, \left(\frac{|\xi|}{r}\right)^{2q}\right\} |\widehat{f}(\xi)|^q \, d\xi\right)^{1/q} \leqslant C\Omega_p[f]\left(\frac{1}{r}\right), \quad 1 < p \leqslant 2,$$

$$\sup_{\xi}\left[\min\left(1, \left(\frac{|\xi|}{r}\right)^2\right)|\widehat{f}(\xi)|\right] \leqslant C\Omega_1[f]\left(\frac{1}{r}\right), \quad p = 1.$$

Just as in the one-dimensional case (Proposition 2), this result immediately implies the following estimates analogous to (1.1).

Corollary 5. *Let $1 \leqslant p \leqslant 2$ and $f \in L^p(R^n)$. Then*

$$\left(\int_{|\xi| \geqslant r} |\widehat{f}(\xi)|^q \, d\xi\right)^{1/q} \leqslant C\Omega_p[f]\left(\frac{1}{r}\right), \quad 1 < p \leqslant 2,$$

$$\sup_{|\xi| \geqslant r} |\widehat{f}(\xi)| \leqslant C\Omega_1[f]\left(\frac{1}{r}\right), \quad p = 1.$$

The proof of Proposition 4 is based on the following estimates for the spherical Bessel function.

Lemma 6. *Let $\alpha > -1/2$ and set $I(\lambda/r, z) = 1 - j_\alpha(\lambda z/r)$. Then*

$$C_{1,\alpha} \min\left\{1, \left(\frac{\lambda}{r}\right)^2\right\} \leqslant \int_0^1 I\left(\frac{\lambda}{r}, z\right) dz \leqslant \sup_{0 \leqslant z \leqslant 1} I\left(\frac{\lambda}{r}, z\right) \leqslant C_{2,\alpha} \min\left\{1, \left(\frac{\lambda}{r}\right)^2\right\}, \tag{2.6}$$

where $C_{1,\alpha}$, $C_{2,\alpha}$ are positive constants.

Proof. The middle inequality is immediate. We first consider the case $0 \leqslant \lambda \leqslant r$ noting that j_α is an even function. The Mehler type representation for j_α is given by:

$$j_\alpha(\lambda x) = \frac{2\Gamma(\alpha+1)x^{-2\alpha}}{\sqrt{\pi}\,\Gamma(\alpha+(1/2))} \int_0^x \left(x^2 - y^2\right)^{\alpha-(1/2)} \cos \lambda y \, dy. \tag{2.7}$$

This could also be called a Poisson type representation, see [14, p. 114]. It immediately gives

$$1 - j_\alpha(\lambda x) = \frac{2\Gamma(\alpha+1)x^{-2\alpha}}{\sqrt{\pi}\,\Gamma(\alpha+(1/2))} \int_0^x \left(x^2 - y^2\right)^{\alpha-(1/2)} (1 - \cos \lambda y) \, dy$$

$$= \frac{4\Gamma(\alpha+1)x^{-2\alpha}}{\sqrt{\pi}\,\Gamma(\alpha+(1/2))} \int_0^x \left(x^2 - y^2\right)^{\alpha-(1/2)} \sin^2(\lambda y/2) \, dy$$

and by a change of variable,

$$I\left(\frac{\lambda}{r},z\right)=\frac{4\Gamma(\alpha+1)}{\sqrt{\pi}\Gamma(\alpha+(1/2))}\int_0^1\left(1-t^2\right)^{\alpha-(1/2)}\sin^2\left(\frac{\lambda zt}{2r}\right)dt.$$

If $u\leqslant\pi/2$, we have $2u/\pi\leqslant\sin u\leqslant u$. Applying this inequality and simplifying the resulting Beta function integral, we obtain for $\lambda/r\leqslant 1$,

$$\frac{1}{\alpha+1}\left(\frac{\lambda z}{\pi r}\right)^2\leqslant I(\lambda/r,z)\leqslant\frac{1}{\alpha+1}\left(\frac{\lambda z}{2r}\right)^2.$$

Integrating, this immediately gives the following for $\lambda/r\leqslant 1$:

$$C_{1,\alpha}\left(\frac{\lambda}{r}\right)^2\leqslant\int_0^1 I\left(\frac{\lambda}{r},z\right)dz\leqslant\sup_{0\leqslant z\leqslant 1}I\left(\frac{\lambda}{r},z\right)\leqslant C_{2,\alpha}\left(\frac{\lambda}{r}\right)^2.$$

In general and with possibly different constant than in the above inequality,

$$\sup_{0\leqslant z\leqslant 1}I\left(\frac{\lambda}{r},z\right)\leqslant C_{2,\alpha}\min\left\{1,\left(\frac{\lambda}{r}\right)^2\right\}$$

since j_α is a bounded function. For the other inequality, if $\lambda/r>1$ we have for fixed $\delta>0$

$$\begin{aligned}\int_0^1 I\left(\frac{\lambda}{r},z\right)dz&=\frac{2\Gamma(\alpha+1)}{\sqrt{\pi}\Gamma(\alpha+(1/2))}\int_0^1\left(1-t^2\right)^{\alpha-(1/2)}\left(1-\frac{\sin(\lambda t/r)}{(\lambda t/r)}\right)dt\\&\geqslant\frac{2\Gamma(\alpha+1)}{\sqrt{\pi}\Gamma(\alpha+(1/2))}\int_\delta^1\left(1-t^2\right)^{\alpha-(1/2)}\left(1-\frac{\sin(\lambda t/r)}{(\lambda t/r)}\right)dt\\&\geqslant C(\delta)\int_\delta^1\left(1-t^2\right)^{\alpha-(1/2)}dt,\end{aligned}$$

where $C(\delta)$ is a positive constant. This concludes the proof. □

Proof of Proposition 4. Starting from the identity

$$\widehat{M^t f}(\xi)=\widehat{f}(\xi)\,j_{\frac{n-2}{2}}\left(t|\xi|\right),$$

we apply the Hausdorff–Young inequality

$$\left\|M^t f-f\right\|_p^q\geqslant C_p\int_{\mathbf{R}^n}\left|\widehat{f}(\xi)\right|^q\left|j_{\frac{n-2}{2}}\left(t|\xi|\right)-1\right|^q d\xi.$$

Taking the supremum, adjusting the variable, and a simple inequality gives:

$$\begin{aligned}\sup_{0<t\leqslant 1/r}\left\|M^t f-f\right\|_p^q &\geqslant C_p \sup_{0\leqslant t\leqslant 1/r}\int_{R^n}\left|\widehat{f}(\xi)\right|^q\left|j_{\frac{n-2}{2}}\left(t|\xi|\right)-1\right|^q d\xi\\ &= C_p \sup_{0\leqslant z\leqslant 1}\int_{\mathbb{R}^n}\left|\widehat{f}(\xi)\right|^q\left|j_{\frac{n-2}{2}}\left(z|\xi|/r\right)-1\right|^q d\xi\\ &\geqslant C_p\int_{R^n}\left|\widehat{f}(\xi)\right|^q\int_0^1\left(1-j_{\frac{n-2}{2}}\left(|\xi|z/r\right)\right)^q dz\,d\xi.\end{aligned}$$

On the compact set $[0,1]$, L^q-norm dominates L^1-norm, hence

$$\begin{aligned}\left(\int_0^1\left(1-j_{\frac{n-2}{2}}\left(z|\xi|/r\right)\right)^q dz\right)^{1/q} &\geqslant \int_0^1\left(1-j_{\frac{n-2}{2}}\left(z|\xi|/r\right)\right)dz\\ &\geqslant C\min\left\{1,\left(|\xi|/r\right)^2\right\},\end{aligned}$$

where we have used Lemma 6. It now follows that

$$\sup_{0\leqslant t\leqslant 1/r}\left\|M^t f-f\right\|_p^q \geqslant C\int_{R^n}\left|\widehat{f}(\xi)\right|^q\min\left\{1,\left(|\xi|/r\right)^{2q}\right\}d\xi,$$

where $C=C(p,n)>0$ as desired. □

In the case where $p=2$, the Hausdorff–Young inequality becomes the Parseval equality. The above methods then carry two-sided estimate as a corollary. For succinct statement, the notation $g(r)\asymp h(r)$ means that g is bounded above and below by a positive constant times h.

Corollary 7. *Let $f\in L^2(\mathbb{R}^n)$. Then*

$$\Omega_2[f]\left(\frac{1}{r}\right)\asymp\left(\int_{\mathbb{R}^n}\left|\widehat{f}(\xi)\right|^2\min\left\{1,\left(\frac{|\xi|}{r}\right)^4\right\}d\xi\right)^{1/2}. \tag{2.8}$$

Call the right-hand side of (2.8) $I(r)$. It can be shown (see [9, Theorem 1.4]) that for some $0<\alpha<2$, $(\int_{|\xi|\geqslant r}|\widehat{f}(\xi)|^2\,d\xi)^{1/2}=O(r^{-\alpha})$ if and only if $I(r)=O(r^{-\alpha})$ as $r\to\infty$. Combining this with the above corollary provides a nice two-way estimate in the L^2-case.

Corollary 8. *Let $f\in L^2(\mathbb{R}^n)$ and let $0<\alpha<2$. Then*

$$\Omega_2[f]\left(\frac{1}{r}\right)\asymp r^{-\alpha}\quad\text{if and only if}\quad\left(\int_{|\xi|\geqslant r}\left|\widehat{f}(\xi)\right|^2 d\xi\right)^{1/2}\asymp r^{-\alpha}.$$

3. Rank one symmetric spaces

Background references for this section are Helgason [10,11] and Koornwinder [12]. Let $X = G/K$ where G is a connected non-compact semisimple Lie group with finite center and real rank one and K is a maximal compact subgroup. On the Lie algebra level we have the Iwasawa decomposition $\mathfrak{g} = \mathfrak{k} + \mathfrak{a} + \mathfrak{n}$, where $\mathfrak{k}$ is the Lie algebra of K and $\mathfrak{a}$ is a maximal abelian subalgebra. The rank one condition is that $\dim \mathfrak{a} = 1$. The nilpotent algebra $\mathfrak{n}$ has root space decomposition $\mathfrak{n} = \mathfrak{n}_\gamma + \mathfrak{n}_{2\gamma}$, where γ and 2γ are the positive roots. Let $m_\gamma = \dim \mathfrak{n}_\gamma$ and $m_{2\gamma} = \dim \mathfrak{n}_{2\gamma}$ be the root space dimensions and set $\rho = \frac{1}{2}(m_\gamma + 2m_{2\gamma})$. Choose $H_0 \in \mathfrak{a}$ such that $\gamma(H_0) = 1$. This allows identifying $\mathfrak{a}$ with $\mathbb{R}$ via $\mathbb{R} \ni t \to tH_0 \in \mathfrak{a}$ and on the dual side, $\mathfrak{a}_{\mathbb{C}}^*$ with $\mathbb{C}$. At the group level, the Iwasawa decomposition has form $G = KAN$ and we write $G \ni g = k(g)\exp(H(g))n(g)$, where $H(g) \in \mathfrak{a}$ and exp is the exponential function.

The polar decomposition of G takes the form $G = KA^+K$, where $A^+ = \{a_t\colon\ t \geqslant 0\}$. Following standard practice, functions f on X are identified with right K-invariant functions on G and write $f(x) = f(g)$, where $x = gK$. Let M be the centralizer of A in K and set $B = K/M$. For $x = gK \in X$ and $b = kM \in B$, set $A(x,b) = -H(g^{-1}k)$. Then the function

$$x \to e^{(i\lambda+\rho)A(x,b)}, \quad \lambda \in \mathbb{C},\ b \in B,$$

is an eigenfunction of the Laplacian on X with eigenvalue $-(\lambda^2 + \rho^2)$. Averaging this function over B leads to the Harish–Chandra formula for the spherical function

$$\phi_\lambda(x) = \int_B e^{(i\lambda+\rho)A(x,b)}\,db, \tag{3.1}$$

where db is normalized measure on B.

The Laplacian on X is denoted L; its radial part is given by

$$L_r = \frac{d^2}{dx^2} + \frac{\Delta'_{(\alpha,\beta)}(x)}{\Delta_{(\alpha,\beta)}(x)}\frac{d}{dx},$$

where $\Delta_{(\alpha,\beta)}(t) = (2\sinh t)^{(2\alpha+1)}(2\cosh t)^{2\beta+1}$, $\alpha = (m_\gamma + m_{2\gamma} - 1)/2$ and $\beta = (m_{2\gamma} - 1)/2$. Notice that $\rho = \alpha + \beta + 1$. The spherical function on X is the unique radial solution to the equation

$$Lu = -\big(\lambda^2 + \rho^2\big)u$$

which is one at the origin of X. Writing $g = ka_t$, it is known [12] that $\phi_\lambda(a_t) = \phi_\lambda^{(\alpha,\beta)}(t)$, Jacobi function of first kind mentioned in the introduction. The Jacobi function admits a Mehler-type representation [12] as follows: let $\alpha > -1/2$ and $-1/2 \leqslant \beta \leqslant \alpha$, then

$$\phi_\lambda^{(\alpha,\beta)}(t) = \frac{2}{\Delta_{(\alpha,\beta)}(t)}\int_0^t \cos\lambda s\, A_{(\alpha,\beta)}(s,t)\,ds, \tag{3.2}$$

where

$$A_{(\alpha,\beta)}(s,t) = \frac{2^{3\alpha+2\beta+\frac{1}{2}}\Gamma(\alpha+1)}{\Gamma(\frac{1}{2})\Gamma(\alpha+\frac{1}{2})}\sinh 2t\cosh^{\beta-\frac{1}{2}} t \times (\cosh t-\cosh s)^{\alpha-\frac{1}{2}} F\left(\frac{1}{2}+\beta;\frac{1}{2}-\beta;\alpha+\frac{1}{2};\frac{\cosh t-\cosh s}{2\cosh t}\right). \quad (3.3)$$

This formula is proved by using an Abel transform which intertwines the Jacobi differential operator and d^2/dx^2. Koornwinder gives geometric perspectives of this in terms of a Radon transform on symmetric space; see also [2]. Given the range for the parameters given above, then key facts known about Jacobi functions, and hence, spherical functions on rank one symmetric spaces are:

- $|\phi^{(\alpha,\beta)}_{\mu+i\eta}(t)| \leqslant \phi^{(\alpha,\beta)}_{i\eta}(t) \leqslant 1$ for $\mu \in \mathbb{R}$ and $|\eta| \leqslant \rho$.
- $|\phi^{(\alpha,\beta)}_{\mu+i\eta}(t)| \leqslant e^{|\eta|t}\phi^{(\alpha,\beta)}_0(t) \leqslant C(1+t)e^{(|\eta|-\rho)t}$.
- Let $1 < p < 2$, and define $D_p = \{\lambda = \mu + i\eta\colon\ |\eta| < (2/p-1)\rho\}$. Then

$$\lambda \in D_p \quad \Rightarrow \quad \phi^{(\alpha,\beta)}_\lambda \in L^q\big(\mathbb{R}^+, \Delta_{(\alpha,\beta)}(t)\,dt\big),$$

  where q is the Holder conjugate index: $1/p + 1/q = 1$.

Koornwinder [12] gives proof of the first two bullet items on the basis of a representation formula for Jacobi functions generalizing (3.1) to the analytic case. They can also be obtained from the Mehler representation (3.2) and properties of the Jacobi differential equation. The third bullet follows immediately from the second and simple computation. All three items are well known in the symmetric space literature [8,11].

We will need the following estimate in the sequel.

Lemma 9. *Let $\alpha > -1/2$, $-1/2 \leqslant \beta \leqslant \alpha$, and let $t_0 > 0$. Then for $|\eta| \leqslant \rho$, there exists a positive constant $C = C(t_0,\alpha,\beta)$ such that*

$$\big|1-\phi^{(\alpha,\beta)}_{\mu+i\eta}(t)\big| \geqslant C\big|1-j_\alpha(\mu t)\big| \quad (3.4)$$

for all $0 \leqslant t \leqslant t_0$. Consequently

$$\int_0^1 \left|1-\phi^{(\alpha,\beta)}_{\mu+i\eta}\left(\frac{z}{r}\right)\right| dz \geqslant C\min\left\{1,\left(\frac{\mu}{r}\right)^2\right\} \quad (3.5)$$

for all $r \geqslant r_0 = t_0^{-1}$.

Proof. Without loss of generality, $t_0 = r_0 = 1$. As the parameters are fixed, write $\Delta(t) = \Delta_{(\alpha,\beta)}(t)$ and $\phi_\lambda(t) = \phi^{(\alpha,\beta)}_\lambda(t)$. Using the Mehler representation (3.2) and the fact that $\phi_{i\rho}(t) = 1$,

$$1-\phi_\lambda(t) = \frac{2}{\Delta(t)}\int_0^t [\cosh\rho s - \cos\lambda s]A(s,t)\,ds.$$

Write $\lambda = \mu + i\eta$, then since the modulus dominates the real part we have,

$$\begin{aligned}\left|1-\phi_{\mu+i\eta}(t)\right| &\geqslant \frac{2}{\Delta(t)}\int_0^t [\cosh\rho s - \cosh\eta s\cos\mu s]A(s,t)\,ds\\ &= \frac{2}{\Delta(t)}\int_0^t \cosh\eta s\left[\frac{\cosh\rho s}{\cosh\eta s} - \cos\mu s\right]A(s,t)\,ds\\ &\geqslant \frac{2}{\Delta(t)}\int_0^t [1-\cos\mu s]A(s,t)\,ds.\end{aligned}$$

Turning attention to the kernel $A(s,t)$ and collecting terms, the right-hand side above is:

$$\frac{2^{3\alpha+2\beta+\frac{1}{2}}\Gamma(\alpha+1)}{\Gamma(\frac{1}{2})\Gamma(\alpha+\frac{1}{2})}\frac{1}{(\sinh t)^{2\alpha}(\cosh t)^{\beta+\frac{1}{2}}}\int_0^t (1-\cos\mu s)(\cosh t-\cosh s)^{\alpha-\frac{1}{2}}G(s,t)\,ds,$$

where $G(s,t)$ is the hypergeometric function in the kernel (3.3). It follows that

$$\begin{aligned}\left|1-\phi_{\mu+i\eta}(t)\right| &\geqslant c(\alpha,\beta)\frac{\Gamma(\alpha+1)}{\Gamma(\frac{1}{2})\Gamma(\alpha+\frac{1}{2})(\sinh t)^{2\alpha}}\\ &\quad\times\int_0^t (1-\cos\mu s)(\cosh t-\cosh s)^{\alpha-\frac{1}{2}}G(s,t)\,ds.\end{aligned}\tag{3.6}$$

Applying a quadratic transformation to the hypergeometric [14, p. 252],

$$G(s,t) = \left(\frac{\cosh t+\cosh s}{2\cosh t}\right)^{\alpha-\frac{1}{2}} F\left(\frac{\alpha-\beta}{2};\frac{\alpha+\beta}{2};\alpha+\frac{1}{2};\frac{\cosh^2 t-\cosh^2 s}{\cosh^2 t}\right).$$

Given the ranges on the parameters and the fact that $s\leqslant t\leqslant 1$, the above hypergeometric can be written as a series of positive terms with leading coefficient one, and hence is greater or equal one. Furthermore, it is easy to see that $(\frac{\cosh t+\cosh s}{2\cosh t})^{\alpha-\frac{1}{2}}\geqslant c>0$ for some constant c. Now for $0\leqslant s\leqslant t\leqslant 1$, using power series it follows that $\frac{1}{2}(t^2-s^2)\leqslant\cosh t-\cosh s\leqslant c'(t^2-s^2)$, for some constant c'. This implies that

$$(\cosh t-\cosh s)^{\alpha-\frac{1}{2}}\geqslant c\left(t^2-s^2\right)^{\alpha-\frac{1}{2}}$$

for a positive constant c. Returning to (3.6), the estimate can now be written

$$\left|1-\phi_{\mu+i\eta}(t)\right|\geqslant c(\alpha,\beta)\frac{\Gamma(\alpha+1)}{\Gamma(\frac{1}{2})\Gamma(\alpha+\frac{1}{2})(\sinh t)^{2\alpha}}\int_0^t (1-\cos\mu s)\left(t^2-s^2\right)^{\alpha-\frac{1}{2}}\,ds.$$

Using the Mehler representation for the spherical Bessel function (2.7) this becomes

$$\left|1-\phi_{\mu+i\eta}(t)\right| \geqslant c(\alpha,\beta)\left(\frac{t}{\sinh t}\right)^{2\alpha}\left[1-j_\alpha(\mu t)\right] \tag{3.7}$$

and the estimate (3.4) follows. The second estimate (3.5) also follows from this by Lemma 6. This concludes the proof. □

In the paper [18], Stanton and Tomas proved a precise asymptotic estimate for spherical functions on rank one symmetric spaces of form $\phi_\lambda(a_t) = j_{\frac{n-2}{2}}(\lambda t) + O(t^2)$ for $\lambda t < 1$ with a different error estimate in the other region. Our lower bound estimate (3.4) is akin to their result although, even in the case $\lambda \in \mathbb{R}$, (3.4) cannot be proved from their asymptotic because the main term and the error have the same order of magnitude.

In contrast to the Bessel function case (2.6), the estimate for Jacobi functions (3.5) is one-sided only. It is clear that an estimate of this form cannot be two-sided since $\phi_0(t) \neq 1$ for $t \neq 0$. However, in the case of real λ the following substitute holds, a result useful in the L^2-estimates below.

Lemma 10. *Let $\alpha > -1/2$, $1/2 \leqslant \beta \leqslant \alpha$, and $t_0 > 0$. Then for $\lambda \in \mathbb{R}$ and $0 \leqslant t \leqslant t_0$, and writing $\phi_\lambda = \phi_\lambda^{(\alpha,\beta)}$,*

$$\phi_0(t) - \phi_\lambda(t) \asymp \left(\frac{t}{\sinh t}\right)^{2\alpha}\left[1-j_\alpha(\lambda t)\right]. \tag{3.8}$$

Moreover, for $r \geqslant r_0 = t_0^{-1}$,

$$\begin{aligned} c_1 \min\left\{1,\left(\frac{\lambda}{r}\right)^2\right\} &\leqslant \int_0^1 \left[\phi_0\left(\frac{z}{r}\right) - \phi_\lambda\left(\frac{z}{r}\right)\right] dz \\ &\leqslant \sup_{0\leqslant z\leqslant 1}\left[\phi_0\left(\frac{z}{r}\right) - \phi_\lambda\left(\frac{z}{r}\right)\right] \leqslant c_2 \min\left\{1,\left(\frac{\lambda}{r}\right)^2\right\}, \end{aligned} \tag{3.9}$$

where $c_k = c_k(r_0,\alpha,\beta) > 0$, $k = 1, 2$.

Proof. The proof is sketched being very similar to that of Lemma 9. Using the Mehler representation (3.2),

$$\phi_0(t) - \phi_\lambda(t) = \frac{2}{\Delta(t)}\int_0^t [1-\cos\lambda s]A(s,t)\,ds.$$

The lower bound estimate in (3.8) follows from the proof of Lemma 3.1. The upper bound estimate follows from the fact that the hypergeometric function in the kernel $A(s,t)$ is bounded above and ideas used in the previous proof. The second part of the lemma is then a consequence of Lemma 6. □

In contrast to the Bessel function estimate (2.6), the estimates (3.5) and (3.9) are for "large" r. The underlying analytical reason for this is the presence of the factors such as $(t/\sinh t)^{2\alpha}$ in (3.7). Geometrically, in the symmetric space setting we are comparing the size of the spherical function on a curved manifold with the same on its flat tangent space. The effects of curvature are well known, e.g., the volume of balls in Euclidean space are essentially powers in the radius where as in rank one symmetric spaces they are exponential.

We now return to discussion of rank one symmetric spaces. The (Helgason) Fourier transform on X is defined by

$$\widehat{f}(\lambda, b) = \int_X f(x) e^{(-i\lambda+\rho)A(x,b)}\, dx, \tag{3.10}$$

where $b \in B = K/M$ and dx is Riemannian measure on X. Helgason [11] has proved an inversion formula, as well as analogs of the Plancherel and Paley–Wiener theorems for this transform. For our purpose, estimates are needed of the Fourier transform of functions which are p-integrable with respect to Riemannian measure. Let $p \geqslant 1$, then $L^p(X)$ is the Banach space of measurable functions on X with finite norm

$$\|f\|_p = \left(\int_X \left| f(x) \right|^p dx \right)^{1/p} < \infty.$$

The following result is key to obtaining estimates; it is essentially contained in [17].

Proposition 11. *Let $1 \leqslant p < 2$ and $f \in L^p(X)$. Then for $\lambda = \mu + i\eta \in D_p$, $\widehat{f}(\lambda, b)$ is defined a.e. (b) and*

$$\int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant c_p\big(|\eta|\big) \|f\|_p, \tag{3.11}$$

where $c_p(s)$ is a positive function defined on $[0, (\frac{2}{p} - 1)\rho)$.

Proof. The Fourier kernel has modulus $|e^{(-i\lambda+\rho)A(x,b)}| = e^{(\eta+\rho)A(x,b)}$ and hence,

$$\int_B \left| e^{(-i\lambda+\rho)A(x,b)} \right| db = \phi_{i\eta}(x).$$

Using the definition and proceeding formally via Tonelli's theorem gives

$$\int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant \int_X \left| f(x) \right| \phi_{i\eta}(x)\, dx.$$

The desired estimate is then obtained by application of Holder's inequality

$$\int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant \|f\|_p \|\phi_{i\eta}\|_q = c_p\big(|\eta|\big) \|f\|_p.$$

Notice that for the right-hand side to be finite, η must satisfy $|\eta| < (\frac{2}{p} - 1)\rho$. This shows that the integral on the right must be finite for $f \in L^p(X)$ and hence, the Fourier transform is defined by an absolutely convergent integral a.e. (b). □

Group theoretically, the spherical mean operator on X is defined by

$$M^t f(g) = \int_K f(gka_t)\,dk.$$

Under the Fourier transform an operational property easily derived from the definitions is

$$\widehat{M^t f}(\lambda, b) = \widehat{f}(\lambda, b)\phi_\lambda(a_t). \tag{3.12}$$

As in the Euclidean case, we define the spherical modulus of continuity as

$$\Omega_p[f](r) = \sup_{0<t\leqslant r} \left\| M^t f - f \right\|_p.$$

Theorem 12. *Let $1 \leqslant p < 2$ and $f \in L^p(X)$. Then for $|\eta| < (2/p - 1)\rho$ and $r \geqslant r_0 > 0$,*

$$\sup_\mu \left[\min\left\{ 1, \left(\frac{\mu}{r}\right)^2 \right\} \int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \right] \leqslant c_p\big(|\eta|\big)\Omega_p[f]\left(\frac{1}{r}\right),$$

where $c_p(|\eta|) > 0$.

Proof. The proof proceeds as in the Euclidean space where the estimate given in Proposition 11 is used instead of the Hausdorff–Young theorem. Specifically from this estimate and the operational property (3.12),

$$\left|\phi_{\mu+i\eta}(a_t) - 1\right| \int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant c_p\big(|\eta|\big) \left\| M^t f - f \right\|_p.$$

Applying the supremum for $t \leqslant 1/r$,

$$\sup_{0\leqslant z\leqslant 1} \left|\phi_{\mu+i\eta}(a_{z/r}) - 1\right| \int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant c_p\big(|\eta|\big)\Omega_p[f](1/r).$$

The result then follows by Lemma 9. □

An immediate corollary is the following estimate in the vein of (1.1) which implies a version of the Riemann–Lebesgue lemma for the Helgason Fourier transform on X.

Corollary 13. *Let $1 \leqslant p < 2$ and $f \in L^p(X)$. Then for $|\eta| < (2/p - 1)\rho$ and $r \geqslant r_0 > 0$,*

$$\sup_{|\mu| \geqslant r} \int_B \left| \widehat{f}(\mu + i\eta, b) \right| db \leqslant c_p\big(|\eta|\big) \Omega_p[f](1/r). \tag{3.13}$$

In particular,

$$\lim_{|\mu| \to \infty} \int_B \left| \widehat{f}(\mu + i\eta, b) \right| db = 0 \tag{3.14}$$

uniformly on the set $|\eta| \leqslant \eta_0 < (\frac{2}{p} - 1)\rho$.

The form of the Riemann–Lebesgue lemma given in (3.14) was proved in [17] using different methods. Notice that the estimate (3.13) quantifies this result.

This section concludes with two results analogous to Theorem 4 for $L^2(X)$ and a discussion of the case of $L^p(X)$. For this, part of the Plancherel theorem for the Fourier transform as proved by Helgason and stated thus: the Fourier transform has continuous extension to $L^2(X)$ such that for $f \in L^2(X)$,

$$\|f\|_2 = \left(\frac{1}{2\pi} \int_{\mathbb{R}} \int_B \left| \widehat{f}(\lambda, b) \right|^2 db \left| c_X(\lambda) \right|^{-2} d\lambda \right)^{1/2}. \tag{3.15}$$

Here, $c_X(\lambda)$ is the Harish–Chandra c-function; its explicit form and structure are not needed here, see [10,12].

The following result is our extension of Proposition 4 to the rank one symmetric spaces.

Theorem 14. *Let $f \in L^2(X)$. Then for $r \geqslant r_0 > 0$,*

$$\left(\int_{\mathbb{R}} \min\left\{ 1, \left(\frac{\lambda}{r} \right)^4 \right\} \int_B \left| \widehat{f}(\lambda, b) \right|^2 db \left| c_X(\lambda) \right|^{-2} d\lambda \right)^{1/2} \leqslant C \Omega_2[f]\left(\frac{1}{r} \right)$$

for some positive constant $C = C(r_0)$.

The proof of this is similar to that in Euclidean space and makesuse of (3.15) and Lemma 9. As mentioned earlier, one cannot expect to get a two-sided estimate in the previous result for $L^2(X)$ as was given in the Euclidean case. However, for a modified modulus of continuity, a two-sided estimate can be proved in the L^2-case. Specifically, the modified modulus of continuity is defined by

$$\Omega'_p[f](r) = \sup_{t \leqslant r} \left\| M^t f - \phi_0(a_t) f \right\|_p. \tag{3.16}$$

It is easy to see that $\Omega'_p[f](r) \leqslant \Omega_p[f](r) + \|f\|_p \sup_{t \leqslant r}[1 - \phi_0(a_t)]$ and hence, $\lim_{r \to 0} \Omega'_p[f](r) = 0$. It is not unheard of to have ϕ_0 appear in function class definitions on symmetric space, e.g., it appears naturally in the definition of Schwartz space [8]. In the vein of this paper, it is the natural factor which permits two-sided estimates as seen in the following result.

Proposition 15. *Let $f \in L^2(X)$. Then for $r \geqslant r_0 > 0$,*

$$\left(\int_{\mathbb{R}} \min\left\{1, \left(\frac{\lambda}{r}\right)^4\right\} \int_B \left|\widehat{f}(\lambda, b)\right|^2 db \left|c_X(\lambda)\right|^{-2} d\lambda \right)^{1/2} \asymp \Omega_2'[f]\left(\frac{1}{r}\right).$$

Moreover, if $0 < \alpha < 2$, then $\Omega_2'[f](\frac{1}{r}) \asymp r^{-\alpha}$ if and only if

$$\left(\int_{|\lambda| \geqslant r} \int_B \left|\widehat{f}(\lambda, b)\right|^2 db \left|c_X(\lambda)\right|^{-2} d\lambda \right) \asymp r^{-\alpha}.$$

The second part of this result is an analog of Corollary 8and is proved in the same way.

The analog of Theorem 14 for $1 < p < 2$ is incomplete due to the lack of a precise Hausdorff–Young inequality for the Helgason Fourier transform. Eguchi and Kumahara [5] proved a Hausdorff–Young inequality for K-invariant functions on symmetric spaces of any rank. Specifically, and in the rank one case, radial functions $f \in L^p(X)$ have Fourier transforms which satisfy

$$\left(\int_{\mathbb{R}} \left|\widehat{f}(\mu + i\eta)\right|^q \left|c_X(\mu)\right|^{-2} d\mu \right)^{1/q} \leqslant C_p(\eta) \|f\|_p, \tag{3.17}$$

where $\lambda = \mu + i\eta \in D_p$. This was later extended to the case of K-finite functions in [6]. Using a Riesz convexity argument between (3.11) for $p = 1$ and $\eta = 0$ and the L^2-isometry (3.15), Mohanty et al. [15] obtained a Hausdorff–Young type inequality, however this result does not contain (3.17) as a special case. Based on the results of Eguchi et al. and the techniques used in this paper, the following can be proved for radial functions and those of K-finite type, and seems plausible in general.

Conjecture 16. *Let $1 < p < 2$ and $f \in L^p(X)$. Then for $\lambda = \mu + i\eta \in D_p$ and $r \geqslant r_0 > 0$,*

$$\left(\int_{\mathbb{R}} \min\left\{1, \left(\frac{\mu}{r}\right)^{2q}\right\} \int_B \left|\widehat{f}(\mu + i\eta, b)\right|^q db \left|c_X(\mu)\right|^{-2} d\mu \right)^{1/q} \leqslant C_p\big(|\eta|\big) \Omega_p[f]\left(\frac{1}{r}\right),$$

where $C_p(\cdot)$ is a positive function on $[0, (\frac{2}{p} - 1)\rho)$.

4. Variations on the theme

In this section variations of the main results (Proposition 4 and Theorem 12) are sketched for the Fourier–Bessel and Jacobi transforms. In both cases an operator with similar properties as the spherical mean operator on Euclidean and symmetric space is needed. In the works of Chebli [3] and Trimeche [22], such a generalized translation operator has been studied and used to develop harmonic analysis of a general class of singular differential operators on $(0, \infty)$. The Fourier–Bessel and Jacobi operators are special examples of this general theory.

In the Fourier–Bessel case, let $\alpha > -1/2$ and consider the Banach space $L^p_\alpha(\mathbb{R}^+) = L^p(\mathbb{R}^+, x^{2\alpha+1}\,dx)$, with norm

$$\|f\|_{p,\alpha} = \left(\int_0^\infty \left|f(x)\right|^p x^{2\alpha+1}\,dx\right)^{1/p}.$$

The needed operator, denoted T_y, $y \in (0,\infty)$, must be a bounded operator $T_y : L^p_\alpha(\mathbb{R}^+) \to L^p_\alpha(\mathbb{R}^+)$, satisfy $\lim_{y\to 0} T_y f = f$ in the norm of $L^p_\alpha(\mathbb{R}^+)$, and possess the operational property

$$\mathcal{F}_\alpha[T_y f](\lambda) = \mathcal{F}_\alpha f(\lambda)\, j_\alpha(\lambda y). \tag{4.1}$$

The generalized translation operator in the sense of Chebli–Trimeche is then given by

$$\begin{aligned} T_y f(x) &= \frac{\Gamma(\alpha+1)}{\sqrt{\pi}\,\Gamma(\alpha+1/2)} \int_0^\pi f\Big(\sqrt{x^2+y^2-2xy\cos\theta}\Big)\sin^{2\alpha+1}\theta\,d\theta \\ &= \int_0^\infty f(z)W(x,y,z)z^{2\alpha+1}\,dz, \end{aligned}$$

where the kernel is given by

$$W(x,y,z) = \begin{cases} c_\alpha \frac{([(x+y)^2-z^2][z^2-(x-y)^2])^{\alpha-1/2}}{(xyz)^{2\alpha}}, & \text{if } |x-y| < z < x+y, \\ 0, & \text{elsewhere,} \end{cases}$$

$c_\alpha = \frac{2^{1-2\alpha}\Gamma(\alpha+1)}{\sqrt{\pi}\,\Gamma(\alpha+1/2)}$. The first form for $T_y f$ with $\alpha = (n-2)/2$ is precisely the expression for the spherical mean of a radial function on $\mathbb{R}^n$. Using the second form for the translation operator, the operational property (4.1) is a consequence of the well-known product formula for Bessel functions [22].

We introduce the following modulus of continuity on $L^p_\alpha(\mathbb{R}^+)$:

$$\Omega_p[f](r) = \sup_{0<y<r} \|T_y f - f\|_{p,\alpha}.$$

Proposition 17. *Let $1 \leqslant p \leqslant 2$, $\alpha > -1/2$, and $f \in L^p_\alpha(\mathbb{R}^+)$. Then there exists a positive constant $C = C(p,\alpha)$ such that*

$$\left(\int_0^\infty \left|\mathcal{F}_\alpha f(\lambda)\right|^q \min\left\{1, \left(\frac{\lambda}{r}\right)^{2q}\right\}\lambda^{2\alpha+1}\,d\lambda\right)^{1/q} \leqslant C\Omega_p[f]\left(\frac{\lambda}{r}\right), \quad 1 < p \leqslant 2,$$

$$\sup_\lambda\left[\left|\mathcal{F}_\alpha f(\lambda)\right| \min\left\{1, \left(\frac{\lambda}{r}\right)^2\right\}\right] \leqslant C\Omega_1[f]\left(\frac{1}{r}\right), \quad p = 1.$$

The proof of this follows the methods given above in Euclidean space. Analogs of Corollaries 5, 7, and 8 can easily be written down and are left to the reader.

In the Jacobi case, consider the Banach space

$$L^p_{(\alpha,\beta)}\big(\mathbb{R}^+\big) = L^p\big(\mathbb{R}^+, \Delta_{(\alpha,\beta)}(t)\,dt\big),$$
$$\|f\|_{p,(\alpha,\beta)} = \left(\int_0^\infty \big|f(x)\big|^p \Delta_{(\alpha,\beta)}(x)\,dx\right)^{1/p}.$$

The generalized translation operator was defined by Flensted-Jensen and Koornwinder [7] and is given by

$$T_y f(x) = \int_0^\infty f(y) K(x, y, z) \Delta_{(\alpha,\beta)}(z)\,dz$$

with kernel

$$K(x, y, z) = \frac{2^{-2\rho}\Gamma(\alpha+1)(\cosh x \cosh y \cosh z)^{\alpha-\beta-1}}{\Gamma(\frac{1}{2})\Gamma(\alpha+\frac{1}{2})(\sinh x \sinh y \sinh z)^{2\alpha}}\big(1-B^2\big)^{\alpha-\frac{1}{2}}$$
$$\times F\left(\alpha+\beta; \alpha-\beta; \alpha+\frac{1}{2}; \frac{1}{2}(1-B)\right)$$

for $|x-y| < z < x+y$ and $K(x, y, z) = 0$ elsewhere and

$$B = \frac{\cosh^2 x + \cosh^2 y + \cosh^2 z - 1}{2\cosh x \cosh y \cosh z}.$$

Koornwinder demonstrates that in the case the parameters correspond to a rank one symmetric space of non-compact type, then $T_y f$ is the spherical mean of a radial function f. The operational property (4.1) then reads

$$\mathcal{F}_{(\alpha,\beta)}[T_y f](\lambda) = \mathcal{F}_{(\alpha,\beta)} f(\lambda)\phi_\lambda^{(\alpha,\beta)}(y).$$

Here, the natural L^p-modulus of continuity is defined analogously to the Fourier–Bessel case, i.e.,

$$\Omega_p[f](r) = \sup_{0<y\leqslant r} \|T_y f - f\|_{p,(\alpha,\beta)}$$

and the extension of Theorem 12 is as follows.

Proposition 18. *Let $\alpha > -1/2$, $-1/2 \leqslant \beta \leqslant \alpha$, and $f \in L^p_{(\alpha,\beta)}(\mathbb{R}^+)$. Then for $|\eta| < (\frac{2}{p} - 1)\rho$ and $r \geqslant r_0 > 0$,*

$$\sup_{\mu}\left[\min\left\{1, \left(\frac{\mu}{r}\right)^2\right\}\left|\mathcal{F}_{(\alpha,\beta)} f(\mu + i\eta)\right|\right] \leqslant c\big(|\eta|\big)\Omega_p[f]\left(\frac{1}{r}\right),$$

where $c(u)$ is a function defined on $[0, (\frac{2}{p} - 1)\rho)$.

For the choice of parameters (α, β) corresponding to a rank one symmetric space of noncompact type, the above result is a specialization of Theorem 12 to radial functions. However, the Jacobi transform appears naturally in other geometric settings, e.g., generalized Gelfand pairs [12] and harmonic AN-groups [1]. Hence, Proposition 18 has interpretation in those settings and it seems likely that the results of Section 3 should have extension, topics we leave for later work.

In closing, it would be interesting to determine a broad subclass of the differential operators considered by Chebli and Trimeche [3,22] for which analogs of Propositions 17 and 18 can be proved. This would bring our results in these examples fully within the framework of harmonic analysis of singular differential operators on the half-line.

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